

SANDAKAN, Malaysia

WMO: 964910

Lat: 5.901N

Lon: 118.059E

Elev: 46

StdP: 14.67

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	73.3	73.9	70.6	113.0	80.4	71.4	116.1	79.7	18.2	81.2	16.4	81.6	4.2	260	0.349

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.8	93.1	80.0	91.7	79.8	91.0	79.8	82.1	88.6	81.6	88.1	81.0	87.5	5.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.5	158.7	86.8	79.5	153.2	85.5	79.1	151.5	85.1	46.1	88.6	45.3	89.0	44.9	88.5	93.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.9	12.1	11.3	DB	71.0	96.2	2.4	2.9	69.3	98.3	67.9	100.0	66.5	101.6	64.8	103.7	(4)
(5)			WB	69.7	84.3	2.1	2.4	68.2	86.1	66.9	87.5	65.7	88.8	64.2	90.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.2	80.5	80.8	81.7	82.8	83.7	83.2	83.0	83.2	82.9	82.3	81.7	81.2	(6)	
	DBStd	1.95	1.92	1.86	1.60	1.46	1.55	1.66	1.75	1.68	1.60	1.52	1.46	1.78	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD50	11770	944	861	984	985	1045	997	1022	1029	987	1000	950	966	(10)	
	CDD65	6295	479	441	519	535	580	547	557	564	537	535	500	501	(11)	
	CDH74	67625	4597	4415	5589	6242	6823	6131	6060	6132	5748	5668	5169	5051	(12)	
	CDH80	23674	1087	1109	1768	2376	2797	2430	2367	2452	2235	2012	1625	1416	(13)	
(14) Wind	WSAvg	4.9	6.4	6.7	6.0	4.9	4.4	4.1	4.1	4.2	4.1	4.1	4.3	5.2	(14)	
(15) Precipitation	PrecAvg	126.9	19.9	11.5	8.0	5.5	5.9	7.4	7.9	10.0	9.6	8.9	13.3	19.0	(15)	
	PrecMax	162.6	49.4	28.0	20.9	13.7	13.7	13.3	17.2	18.8	20.4	14.0	32.1	34.8	(16)	
	PrecMin	99.0	4.4	1.9	1.8	0.6	2.1	2.2	2.8	3.3	3.4	3.6	6.9	5.8	(17)	
	PrecStd	21.1	12.5	7.4	5.5	4.1	3.3	3.1	4.0	4.0	5.0	3.1	5.6	8.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.2	88.1	89.6	91.7	93.5	93.4	93.5	93.6	93.4	92.4	91.1	89.8	(19)	
		MCWB	79.6	79.2	79.8	80.7	81.0	80.2	79.9	79.5	79.6	79.8	79.3	79.8	(20)	
	2%	DB	86.3	86.6	88.2	90.0	91.6	91.7	91.7	92.4	91.6	90.0	88.9	88.0	(21)	
		MCWB	79.2	78.8	79.5	80.2	80.7	80.0	79.6	79.4	79.5	79.6	79.6	80.1	(22)	
	5%	DB	85.8	86.1	87.6	89.4	90.3	90.3	90.3	91.0	90.0	88.9	87.8	86.8	(23)	
		MCWB	79.0	78.6	79.3	80.0	80.5	79.9	79.5	79.2	79.5	79.6	79.9	79.7	(24)	
	10%	DB	84.5	85.3	86.4	88.1	89.5	89.3	89.0	89.3	88.8	87.7	86.4	85.9	(25)	
		MCWB	78.3	78.3	78.8	79.8	80.3	79.8	79.5	79.2	79.5	79.6	79.4	79.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.0	80.8	81.1	82.3	82.9	82.6	82.6	82.0	81.9	81.7	81.6	81.7	(27)	
		MCDB	85.5	85.7	86.9	89.0	90.1	89.2	88.5	88.7	88.9	87.9	87.1	86.7	(28)	
	2%	WB	80.1	79.9	80.4	81.4	81.9	81.7	81.3	81.0	81.0	80.9	80.8	80.8	(29)	
		MCDB	84.6	84.9	86.4	87.9	88.9	88.5	88.0	88.2	88.0	87.4	86.4	85.6	(30)	
	5%	WB	79.3	79.2	79.8	80.7	81.4	80.9	80.5	80.3	80.4	80.2	80.2	80.2	(31)	
		MCDB	83.6	84.1	85.5	87.1	88.1	87.8	87.6	87.6	87.4	86.6	85.6	84.8	(32)	
	10%	WB	78.9	78.7	79.2	80.1	80.8	80.2	79.9	79.7	79.8	79.7	79.7	79.6	(33)	
		MCDB	83.2	83.6	84.6	86.1	87.3	86.8	86.6	86.8	86.6	85.8	84.8	84.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	8.7	10.1	11.5	12.8	13.4	13.6	14.2	13.9	12.3	10.8	9.5	(35)	
		MCDBR	9.5	9.6	10.9	12.5	13.8	14.8	15.1	15.9	15.2	13.8	12.3	10.9	(36)	
	5% WB	MCWBR	4.5	4.3	4.5	4.7	5.3	5.8	5.9	6.1	6.1	5.6	5.2	4.8	(37)	
		MCWBR	8.9	9.2	10.4	11.8	13.1	14.0	14.2	14.8	14.4	12.8	11.3	10.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.425	0.424	0.419	0.420	0.406	0.406	0.410	0.436	0.446	0.453	0.414	0.414	(40)	
		taud	2.462	2.441	2.453	2.442	2.506	2.513	2.501	2.403	2.369	2.347	2.483	2.497	(41)	
	Ebn at Noon		284	286	287	282	280	276	276	273	275	274	285	285	(42)	
		Edh at Noon	36	38	37	37	34	33	34	38	40	41	35	35	(43)	
(44) All-Sky Solar Radiation	RadAvg		1340	1540	1708	1801	1763	1644	1648	1703	1683	1572	1441	1361	(44)	
	RadStd		137	148	143	112	105	84	89	84	103	79	99	131	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.54	+0.68	+0.52	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.23	N/A	N/A	+0.49	+0.61	+0.45	N/A	N/A	+169	+169

Nomenclature: See separate page